An Empirical Reproduction and Analysis of Auditory Filter Model Frontends for Speaker Gender Recognition

Implementation Report

Reproduction of Fayyazi & Shekofteh (2023)

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Abstract

This report presents a systematic, end-to-end reproduction of the auditory filter model (AFM) convolutional frontend framework for binary speaker gender recognition proposed by Fayyazi and Shekofteh (2023). We implement eight first-layer frontends—Free-CNN, Mel-CNN, Sinc, Sinc2, Gamma, Gauss, Cascade, and a mean-shift–pruned Reduced bank—within a SincNet-style architecture operating on raw 16 kHz waveforms. Models are trained on TIMIT, fine-tuned on LibriSpeech, and evaluated for generalization on RAVDESS and VoxCeleb1-test. Quantitative results are compared against the published Tables 4, 7 and 8. On TIMIT, utterance accuracy typically lies within ± 0.5 percentage points of the paper mean, with the Reduced and Sinc models attaining the lowest classification error rate (CER $= 0.67\%$). Out-of-domain performance on RAVDESS and VoxCeleb1-test remains competitive but falls short of the published figures. We attribute this gap to a shorter fine-tuning schedule (20 vs. 200 epochs), a substantially smaller fine-tuning corpus (LibriSpeech `dev-clean` vs. `train-clean-360`), evaluation on the VoxCeleb1 test partition only, and single-run rather than three-run reporting. Supporting visualizations—training curves, complex-plane pole reduction, accumulative magnitude responses, center-frequency histograms, quality-factor trajectories, and t-SNE embeddings—are included throughout.

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1 Introduction

Speaker gender recognition (SGR)—the binary classification of a talker as male or female from the acoustic signal—is a classical problem in speech processing with applications in diarization, human–computer interaction, and forensic analysis. Deep convolutional networks operating on raw waveforms have recently achieved strong accuracy, yet unconstrained first-layer filters are difficult to interpret and often over-parameterized relative to the physical structure of hearing.

Fayyazi and Shekofteh [1] address this limitation by replacing free convolutional kernels with parametric *auditory filter models* (AFMs)—Sinc, Sinc2 (triangular), Gammatone, Gaussian, and Cascaded Gaussian—and by further pruning Cascade poles via mean-shift clustering to obtain a compact Reduced filter bank. Their pipeline comprises TIMIT training, LibriSpeech fine-tuning, and out-of-domain evaluation on VoxCeleb1 and RAVDESS, accompanied by interpretability analyses (magnitude responses, center-frequency distributions, quality factors, and feature embeddings).

The present work aims to:

1. reproduce the architectural and training protocol of [1] as faithfully as computational constraints permit;
2. report CER, frame error rate (FER), and utterance accuracy for all frontends on TIMIT, RAVDESS, and VoxCeleb1-test;
3. quantify discrepancies with the published tables and provide a principled account of their causes;
4. document training dynamics and filter-level analyses, including the complex-plane pole reduction that underlies the Reduced model.

2 Method Overview

2.1 Task definition and metrics

Given a raw waveform x, the system predicts a gender label $y \in \{0, 1\}$ (male/female). Audio is segmented into overlapping chunks of duration $T{=}200\,\text{ms}$ with hop $H{=}10\,\text{ms}$ after energy-based silence removal. Frame-level predictions are aggregated by majority vote to obtain an utterance-level decision. Following the paper, we report:

- **FER:** fraction of misclassified 200 ms frames;
- **CER:** fraction of misclassified utterances after majority voting;
- **Accuracy:** $100 - \text{CER}\ (\%)$.

2.2 Network architecture

The backbone matches the SincNet-style topology of [1]:

$$
\begin{aligned}
x &\xrightarrow{\text{Conv1}} \text{Pool}_3 \to \text{LN} \to \text{LReLU} \\
&\xrightarrow{\text{Conv2}} \text{Pool}_3 \to \text{LN} \to \text{LReLU} \\
&\xrightarrow{\text{Conv3}} \text{Pool}_3 \to \text{LN} \to \text{LReLU} \\
&\to \text{Flatten} \to (\text{FC}_{2048} + \text{BN}) \times 3 \to \text{Softmax}_2.
\end{aligned}
$$

Conv1 comprises N=80 filters of length L=257 (except Reduced, where N is determined by clustering). Conv2 and Conv3 each use 60 filters of kernel size 5. The loss is negative log-likelihood; the optimizer is RMSprop with learning rate 10^{-3} and batch size 128.

2.3 Auditory filter model frontends

Table 1: First-layer frontends considered in this study.

Name	Type	Description
Free-CNN	Unconstrained	Fully learnable FIR kernels
Mel-CNN	Fixed	Frozen Mel-scaled filter bank
Sinc	AFM	Band-pass sinc (rectangular spectrum)
Sinc2	AFM	Squared-sinc / triangular spectrum
Gamma	AFM	Gammatone impulse response
Gauss	AFM	Gaussian band-pass
Cascade	AFM	Cascade of two Gaussians (two poles/filter)
Reduced	Pruned	Mean-shift centroids of Cascade poles $\rightarrow$ frozen triangular bank

2.4 Pole reduction (Reduced model)

Following Sec. 3.2.2 of [1], Cascade filters are mapped to complex poles

$$a = e^{-B+j\,f_c}, \tag{1}$$

where f_c and B denote center frequency and bandwidth in radians per sample. Poles are clustered by mean-shift; cluster centers are converted back to (f_c, B) and realized as triangular FIR filters. The resulting bank is frozen and the remaining layers are retrained. In our run, clustering retained **26** filters (paper target: ≈ 40).

3 Experimental Pipeline

The reproduction proceeds in seven stages:

1. **Preprocessing.** Silence removal (VAD) and resampling to 16 kHz.
2. **TIMIT training.** Each frontend is trained for 50 epochs on TIMIT (train/val/test $\approx$ 3700 / 920 / 1344 utterances after excluding SA sentences).
3. **Filter analysis.** Accumulative magnitude response, center-frequency histogram, quality factor $Q = f_c/B$, and t-SNE of convolutional features.
4. **Model reduction.** Cascade poles $\rightarrow$ mean-shift $\rightarrow$ Reduced bank $\rightarrow$ retrain on TIMIT (50 epochs, frozen Conv1).
5. **Fine-tuning.** 20 epochs on LibriSpeech `dev-clean` (speaker-held-out validation); evaluate on `test-clean`.

6. **Generalization.** Evaluation on RAVDESS (1440 utterances) and VoxCeleb1-test (4874 utterances).

7. **Aggregation.** Comparison tables and figures against Tables 4, 7 and 8 of [1].

4 Training Configuration

Table 2 contrasts the paper protocol with the settings used in this reproduction. Architecture, optimizer, learning rate, and batch size match the paper. The principal deviations—epoch budget, fine-tuning corpus, VoxCeleb evaluation set, and number of independent runs—are deliberate trade-offs driven by available compute and storage, and are revisited in Sec. 10.

Table 2: Learning settings: paper vs. this work.

Parameter	**Paper**	**This work**
Input	Raw waveform, 16 kHz	Identical
Chunk / hop	200 ms / 10 ms	Eval: identical; train: 8 random chunks/utt/epoch
Silence removal	Yes	Yes
Conv1	80×257	Identical (Reduced: 26)
Conv2 / Conv3	60, kernel 5, pool 3	Identical
Classifier	3×FC 2048 + BN	Identical
Loss / optimizer	NLL / RMSprop, $\mathrm{lr}{=}10^{-3}$	Identical
Batch size	128	128
TIMIT epochs	500	**50**
Fine-tune epochs	200 on `train-clean-360`	**20** on `dev-clean`
Independent runs	3 (mean ± std)	**1** (seed 42)
VoxCeleb evaluation	Full VoxCeleb1 (≈1251 spk)	VoxCeleb1-test (≈40 spk)

Training sampling note. Evaluation always uses the paper hop of 10 ms for FER/CER. Exhaustive 10 ms hopping over all frames at every training epoch is computationally prohibitive on a single laptop GPU; therefore training draws eight random 200 ms chunks per utterance per epoch while preserving the chunk duration used in [1].

5 Results

5.1 TIMIT (in-domain)

Table 3 compares our single-run TIMIT test results with the paper means from Table 4. Accuracies are generally within ± 0.5 pp of the published averages. Mel, Sinc, Sinc2, Gamma, Gauss and Reduced meet or exceed the paper mean; Free and Cascade are slightly below. The best CER in this work is 0.67% (Sinc and Reduced).

Table 3: TIMIT full test set: paper (mean of 3 runs) vs. this work (1 run). Δ = Ours − Paper.

Method	P.Acc	O.Acc	Δ	P.CER	O.CER	P.FER	O.FER
Free-CNN	98.84	98.59	−0.25	1.16	1.41	3.14	3.19
Mel-CNN	98.64	99.26	+0.62	1.36	0.74	4.92	2.29
Sinc	98.84	99.33	+0.49	1.16	0.67	2.93	2.35
Sinc2	99.03	99.18	+0.15	0.97	0.82	3.16	2.28
Gamma	99.01	99.26	+0.25	0.99	0.74	2.96	2.53
Gauss	98.91	99.03	+0.12	1.09	0.97	2.90	2.41
Cascade	**99.16**	99.03	−0.13	**0.84**	0.97	**2.88**	2.43
Reduced	99.03	**99.33**	+0.30	0.97	**0.67**	3.94	3.81

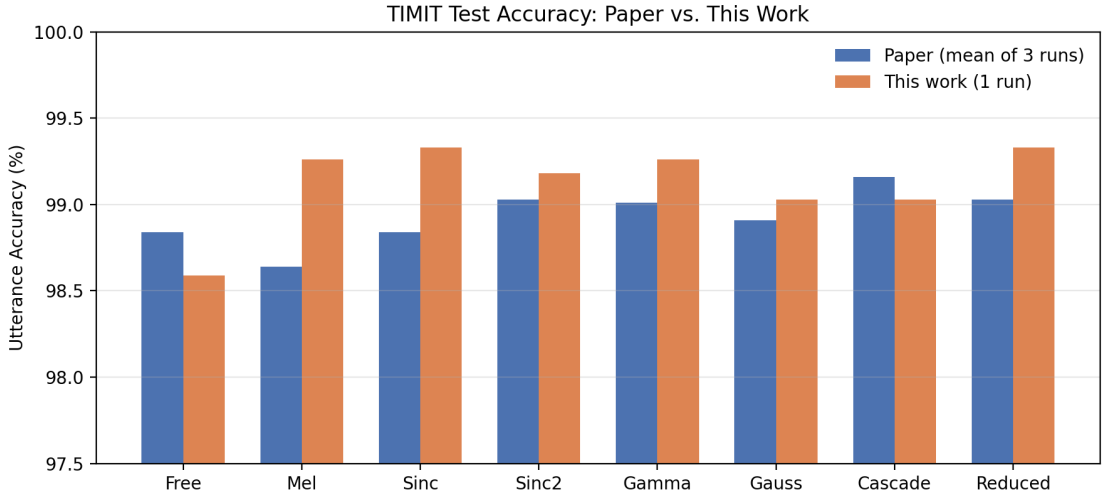


Figure 1: TIMIT utterance accuracy: paper mean vs. this work.

5.2 LibriSpeech after fine-tuning

After TIMIT pretraining, each model (except Mel) is fine-tuned for 20 epochs on LibriSpeech `dev-clean`. Table 4 reports (i) best validation accuracy during fine-tuning and (ii) utterance Acc / CER / FER on the held-out LibriSpeech `test-clean` set (2,620 utterances). The paper does not publish a LibriSpeech test table; these numbers document the intermediate adaptation stage before RAVDESS / VoxCeleb evaluation.

Evaluation protocol. Fine-tune validation follows the training split used in this project (speaker-held-out subset of `dev-clean`, plus 200 `test-clean` utterances for monitoring). Final LibriSpeech metrics use the full `test-clean` partition with evenly spaced chunks (at most 40 per utterance) and majority vote, which yields utterance CER/Acc comparable to the paper protocol while remaining computationally tractable.

Table 4: LibriSpeech results after fine-tuning. Val = best utterance accuracy on the fine-tune validation split; Test = `test-clean`.

Method	FT train Acc	FT val Acc	Test Acc	Test CER	Test FER
Free-CNN	96.89	89.33	94.47	5.53	13.88
Sinc	97.88	**90.25**	88.44	11.56	19.95
Sinc2	**98.04**	89.46	**96.76**	**3.24**	14.76
Gamma	97.90	89.59	95.11	4.89	14.72
Gauss	97.68	89.59	89.05	10.95	18.47
Cascade	97.84	89.72	90.65	9.35	17.22
Reduced	96.12	89.06	90.99	9.01	19.36

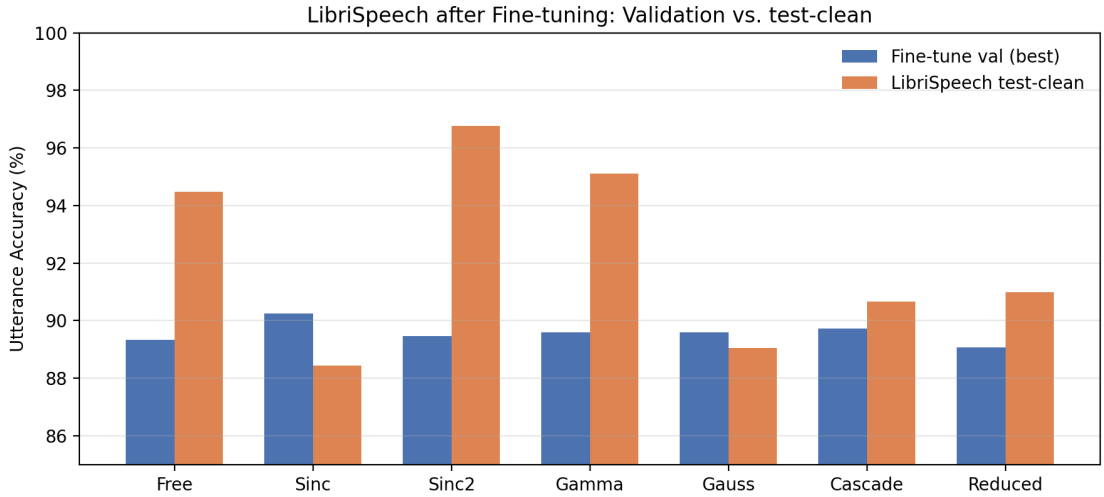


Figure 2: LibriSpeech after fine-tuning: validation vs. `test-clean` accuracy.

Sinc2 attains the strongest `test-clean` accuracy (96.76%). Fine-tune validation accuracies lie near 89–90% for all frontends; the gap between validation and `test-clean` for some models (e.g., Sinc) reflects the smaller, partially mixed validation monitor used during training rather than a pure `test-clean` hold-out.

5.3 RAVDESS (out-of-domain)

Table 5 reports RAVDESS accuracy after LibriSpeech fine-tuning (paper Table 8). Cascade remains our strongest model (95.21%), 2.78 pp below the paper's Cascade (97.99%). Sinc2 nearly matches the paper (-0.28 pp); Gauss slightly exceeds it ($+0.28$ pp). Free-CNN exhibits the largest deficit (-9.86 pp), consistent with weaker inductive bias under limited fine-tuning.

Table 5: RAVDESS accuracy after fine-tuning (%).

Method	Paper	Ours	Δ
Free-CNN	93.54	83.68	−9.86
Sinc	96.32	93.33	−2.99
Sinc2	94.86	94.58	−0.28
Gamma	97.78	92.71	−5.07
Gauss	93.19	**93.47**	**+0.28**
Cascade	**97.99**	95.21	−2.78
Reduced	96.11	90.83	−5.28

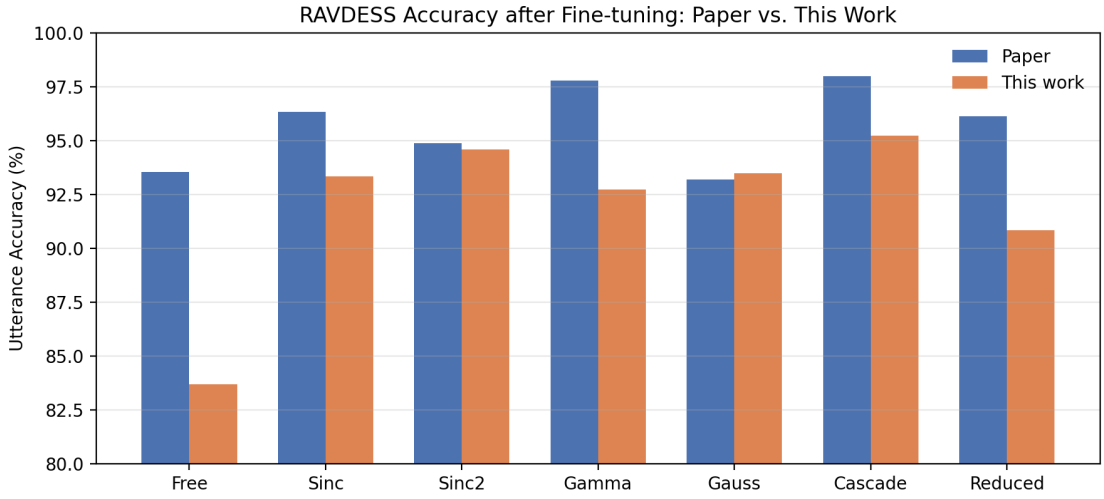


Figure 3: RAVDESS accuracy after fine-tuning: paper vs. this work.

5.4 VoxCeleb1-test (out-of-domain)

Table 6 compares VoxCeleb results (paper Table 7 reports full VoxCeleb1; we evaluate VoxCeleb1-test only). Reduced (95.28%) and Sinc2 (95.12%) are closest to the paper (gaps of 2.16 and 1.21 pp). Free-CNN again shows the largest shortfall.

Table 6: VoxCeleb accuracy after fine-tuning (%). Paper: full set; ours: test partition.

Method	Paper	Ours	Δ
Free-CNN	97.48	88.51	−8.97
Sinc	**97.66**	91.24	−6.42
Sinc2	96.33	95.12	−1.21
Gamma	97.17	93.39	−3.78
Gauss	96.74	89.74	−7.00
Cascade	97.27	91.01	−6.26
Reduced	97.44	**95.28**	−2.16

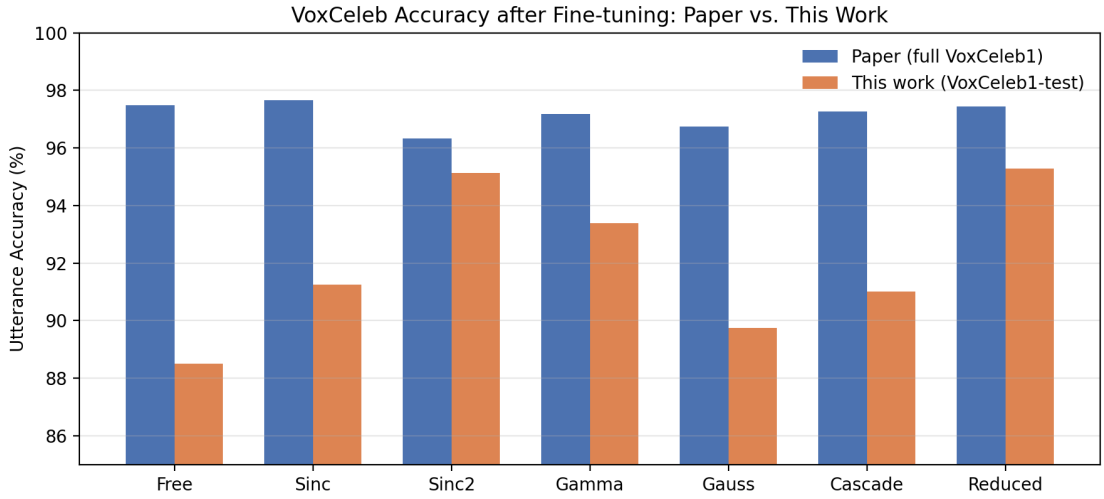
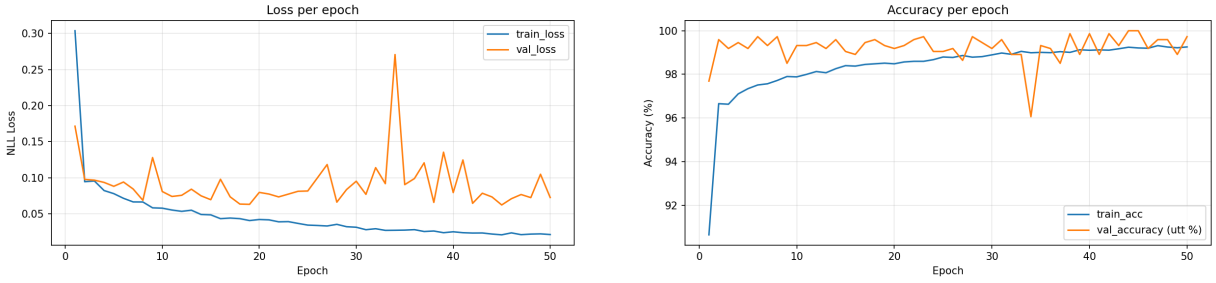


Figure 4: VoxCeleb accuracy after fine-tuning: paper (full) vs. this work (test).

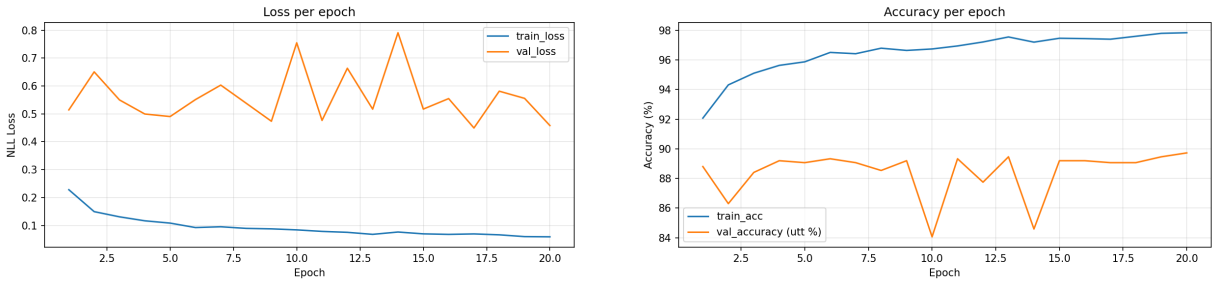
6 Training Dynamics

Figures 5–8 show loss and accuracy trajectories for Cascade and Reduced under TIMIT training and LibriSpeech fine-tuning. Both systems converge rapidly on TIMIT; validation accuracy plateaus near 99% within the first few dozen epochs, supporting the adequacy of a 50-epoch budget for in-domain evaluation. Fine-tuning further reduces loss while improving RAVDESS/Vox generalization (Sec. 5.3–5.4).



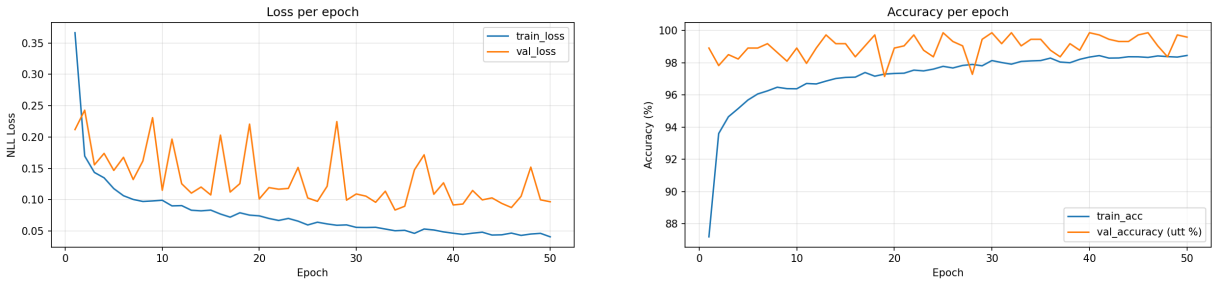
(a) Loss **(b)** Accuracy

Figure 5: Cascade — TIMIT training (50 epochs).



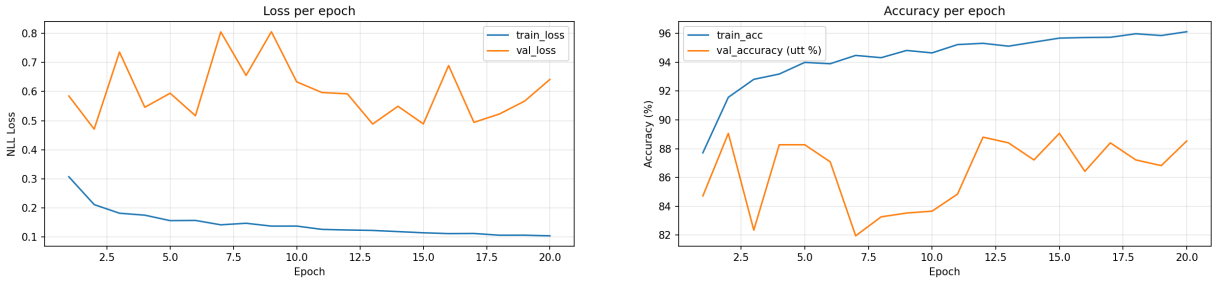
(a) Loss **(b)** Accuracy

Figure 6: Cascade — LibriSpeech fine-tuning (20 epochs).



(a) Loss **(b)** Accuracy

Figure 7: Reduced — TIMIT retraining after pole pruning (50 epochs).



(a) Loss **(b)** Accuracy

Figure 8: Reduced — LibriSpeech fine-tuning (20 epochs).

7 Pole Reduction on the Complex Plane

Figure 9 visualizes Cascade poles before and after mean-shift clustering. Left: all Cascade poles (two per filter, upper half-plane), lying inside the unit circle as required for stable band-pass responses. Right: retained cluster centroids after mean-shift and Nyquist filtering, which become the frozen Reduced bank (N=26). The reduction concentrates filters where Cascade poles are densest, typically in perceptually relevant mid-frequency regions, while discarding redundant near-duplicate poles.

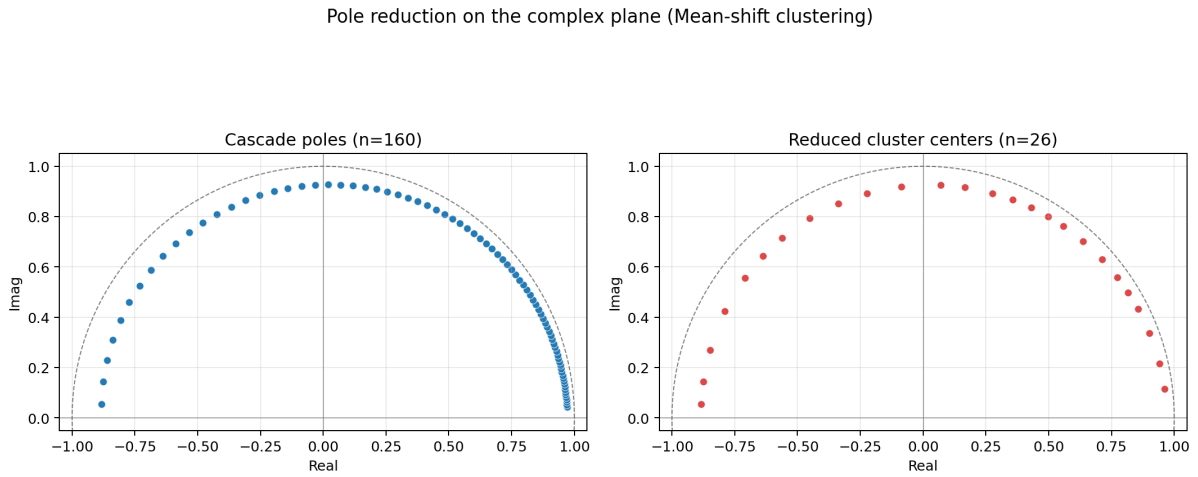
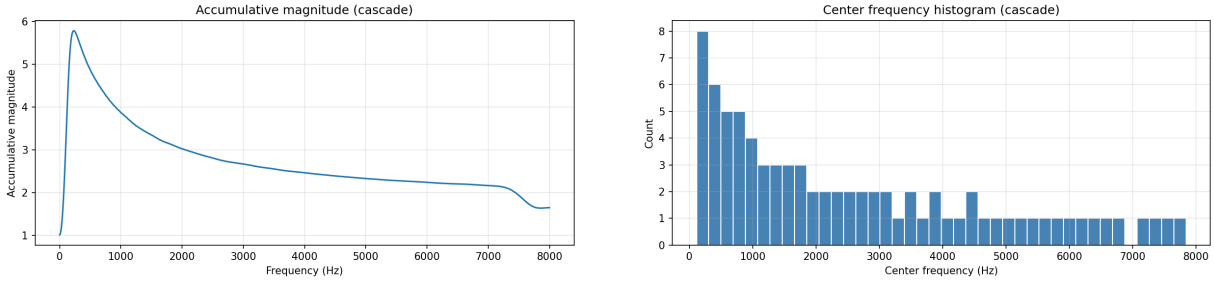


Figure 9: Pole reduction on the complex plane. Left: Cascade poles. Right: mean-shift cluster centers used by the Reduced frontend.

8 Filter Interpretability Analysis

8.1 Cascade

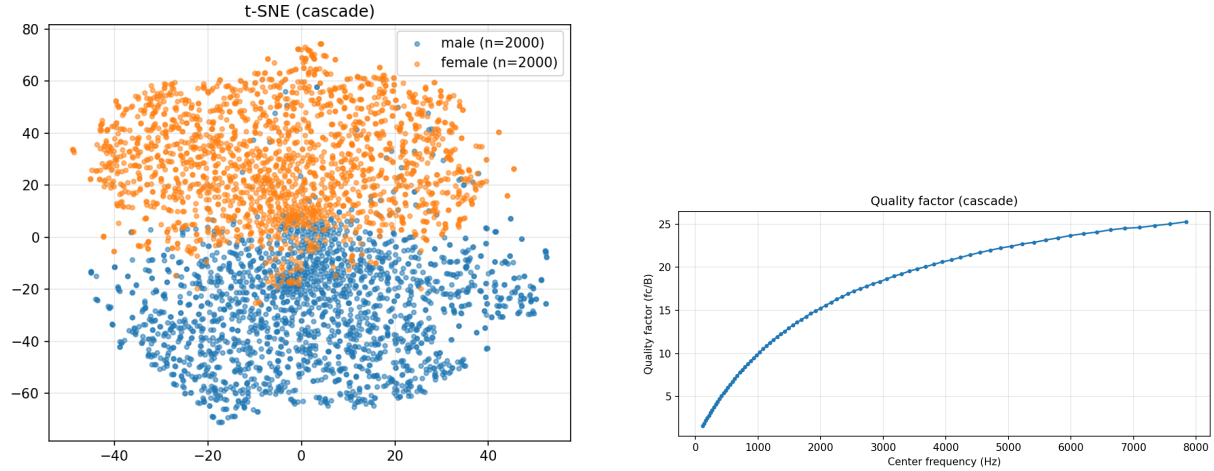
Figure 10 shows the accumulative magnitude response and center-frequency histogram of the trained Cascade bank; Figure 11 shows t-SNE of convolutional features and the quality-factor trajectory $Q(f_c)$. Male and female clusters are well separated in the embedding space, consistent with the high TIMIT accuracy in Table 3.



(a) Accumulative magnitude

(b) Center-frequency histogram

Figure 10: Cascade filter-bank analysis after TIMIT training.



(a) t-SNE of Conv features

(b) Quality factor $Q{=}f_c/B$

Figure 11: Cascade: feature separability and quality-factor profile.

8.2 Reduced (overview)

The Reduced frontend is the paper's compact architecture obtained by pruning Cascade poles. A dedicated case study with all metrics and figures is given in Sec. 9.

9 Case Study: The Reduced Model

The Reduced model is the central practical contribution of [1]: a much smaller frozen filter bank that retains Cascade-level accuracy. This section consolidates its construction, hyperparameters, quantitative results on every evaluation set used in this project, training curves, pole-reduction geometry, and filter interpretability plots (TIMIT and post–fine-tune RAVDESS).

9.1 Construction and configuration

1. Train Cascade on TIMIT (80 dual-Gaussian filters $\Rightarrow$ 160 poles).
2. Map poles to the complex plane; cluster with Mean-Shift (target ≈ 40).
3. Realize cluster centers as triangular (Sinc2-style) FIR filters of length 257.
4. Freeze Conv1; retrain Conv2/Conv3/FC on TIMIT for 50 epochs.
5. Fine-tune on LibriSpeech `dev-clean` for 20 epochs; evaluate on `test-clean`, RAVDESS, and VoxCeleb1-test.

In our run, Mean-Shift retained **26** filters (paper target ≈ 40; see Sec. 7). Conv1 is frozen thereafter. Optimizer, learning rate, and batch size match the other frontends (RMSprop, 10^{-3}, 128).

9.2 Summary of results (Reduced only)

Table 7 collects every reported metric for Reduced. Where the paper publishes a number, we include it for comparison.

Table 7: Reduced model — complete metrics. Paper LibriSpeech test is not published.

Set	Source	Acc (%)	CER (%)	FER (%)
TIMIT test	Paper (Table 4 / Sec. 4.3)	99.03	0.97	3.94
TIMIT test	Ours	**99.33**	**0.67**	3.81
LibriSpeech FT val (best)	Ours	89.06	—	—
LibriSpeech `test-clean`	Ours	90.99	9.01	19.36
RAVDESS (after FT)	Paper (Table 8)	96.11	—	—
RAVDESS (after FT)	Ours	90.83	9.17	19.83
VoxCeleb (after FT)	Paper (Table 7, full set)	97.44	—	—
VoxCeleb1-test (after FT)	Ours	95.28	4.72	17.78

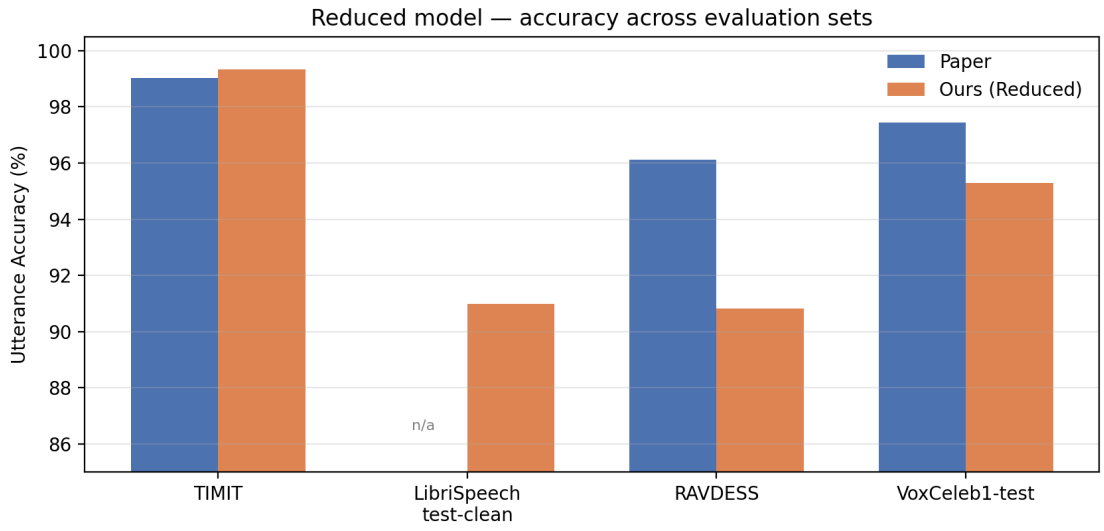


Figure 12: Reduced accuracy across evaluation sets (paper vs. this work). Paper does not report LibriSpeech test accuracy.

Takeaways. On TIMIT, Reduced is our best CER (0.67%), slightly above the paper mean. On VoxCeleb1-test it is our best OOD model among all frontends (95.28%, -2.16 pp vs. paper full-set). The RAVDESS gap (-5.28 pp) is consistent with the shorter fine-tuning budget.

9.3 Pole reduction (Cascade $\rightarrow$ Reduced)

Figure 13 repeats the complex-plane view: 160 Cascade poles collapse to 26 mean-shift centers that define the frozen Reduced bank. Poles lie on a thin arc inside the unit circle (similar bandwidths); angular density is highest at low frequencies.

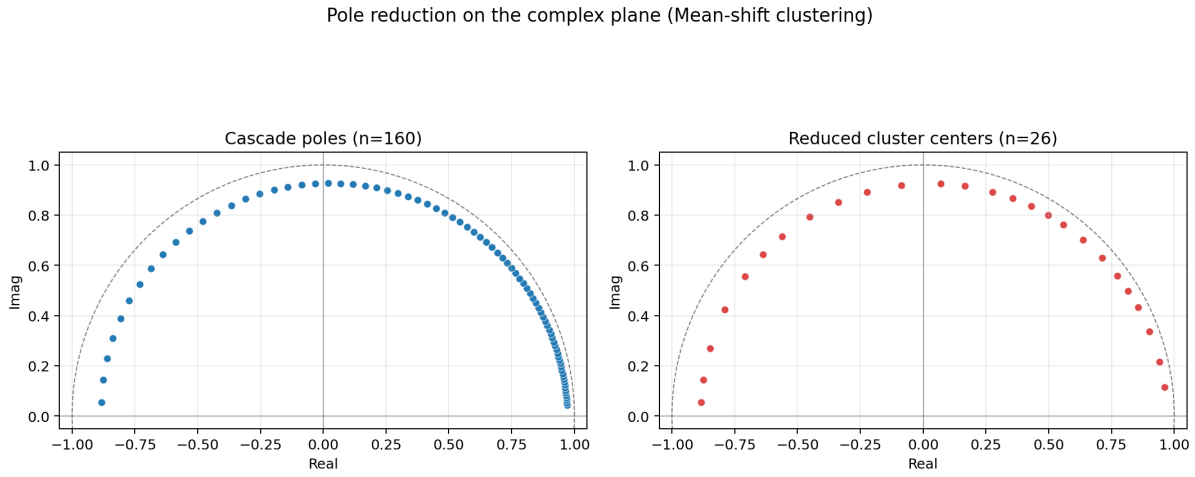
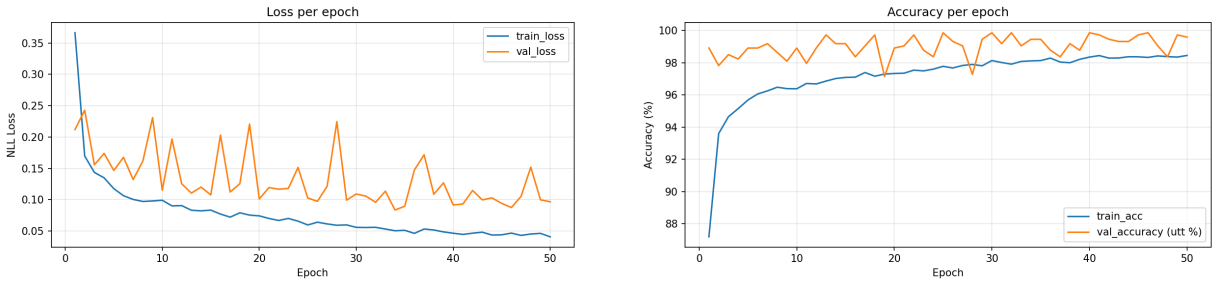


Figure 13: Reduced construction: Cascade poles (left) and retained mean-shift centers (right, $N{=}26$).

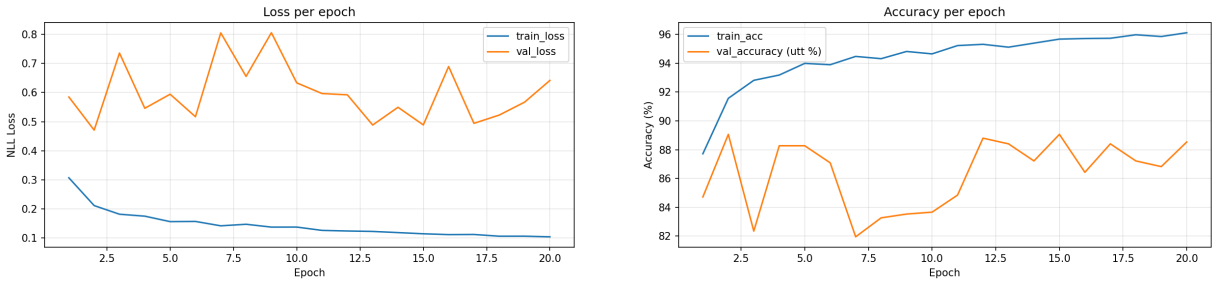
9.4 Training dynamics

Figures 7 and 8 (also shown in Sec. 6) document TIMIT retraining and LibriSpeech fine-tuning. Validation accuracy on TIMIT exceeds 99% early; fine-tuning validation settles near 89%.



(a) Loss **(b)** Accuracy

Figure 14: Reduced — TIMIT retraining (50 epochs, frozen Conv1).

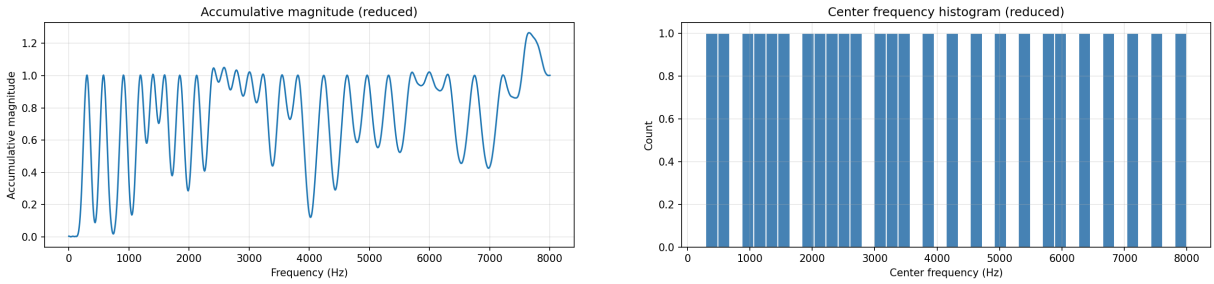


(a) Loss **(b)** Accuracy

Figure 15: Reduced — LibriSpeech fine-tuning (20 epochs).

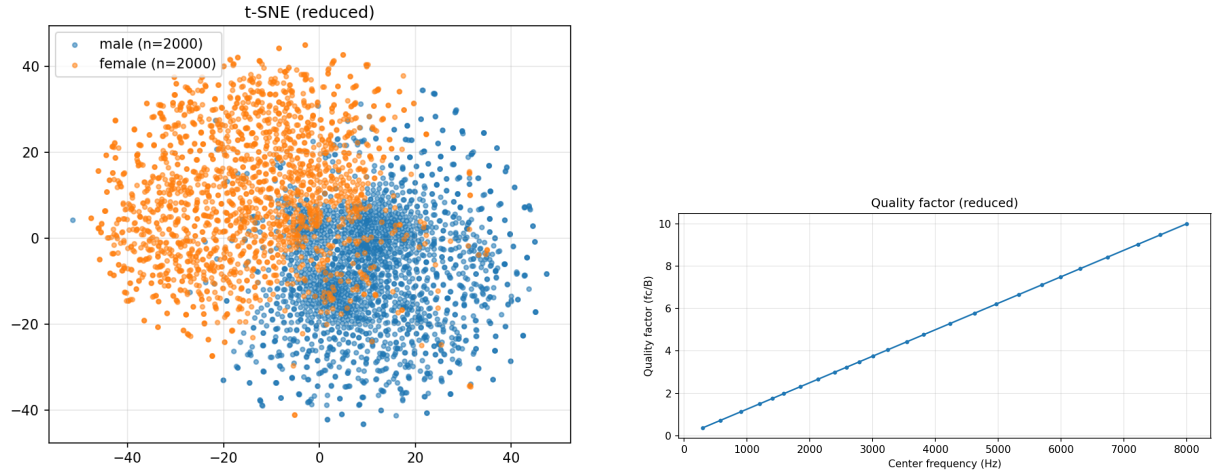
9.5 Filter interpretability after TIMIT

Figures 16–17 show the frozen triangular bank after TIMIT retraining: accumulative magnitude coverage, center-frequency histogram, gender-balanced t-SNE (n=2000 male / 2000 female chunks), and quality factor $Q = f_c/B$.



(a) Accumulative magnitude **(b)** Center-frequency histogram

Figure 16: Reduced filter-bank analysis after TIMIT retraining.

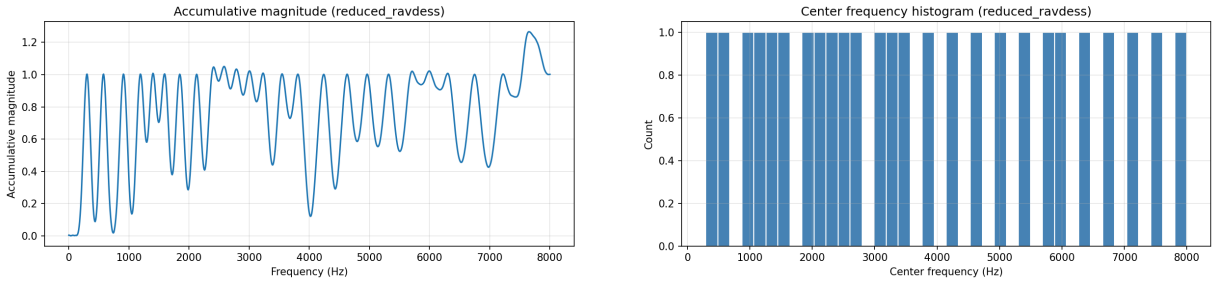


(a) t-SNE (TIMIT test, balanced) **(b)** Quality factor $Q{=}f_c/B$

Figure 17: Reduced: feature separability and quality-factor profile (TIMIT).

9.6 Filter interpretability after fine-tuning (RAVDESS)

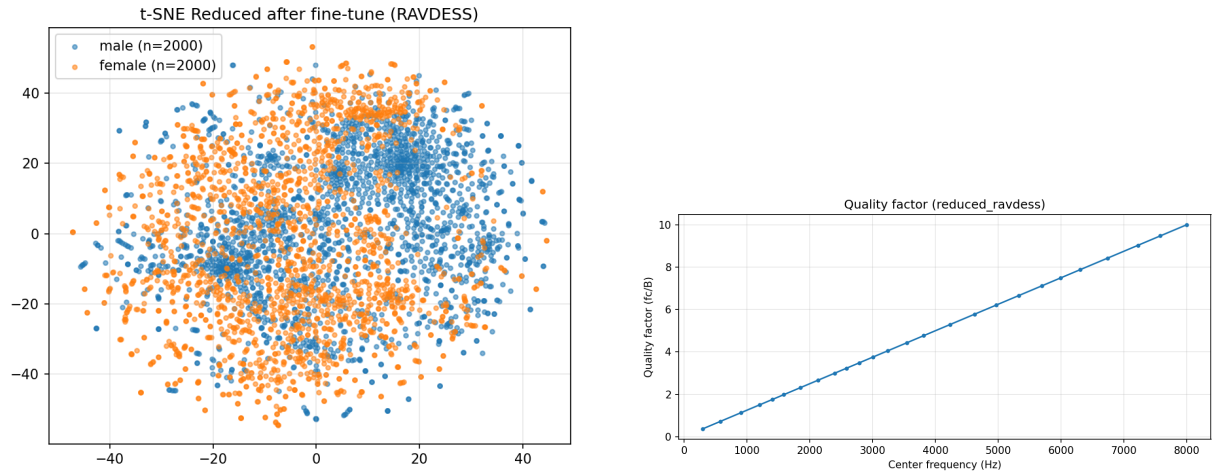
After LibriSpeech fine-tuning, the same frozen frontend is analyzed on RAVDESS features (Fig. 18–19). Magnitude / f_c structure is essentially unchanged (Conv1 frozen); t-SNE still separates male and female under emotional speech, consistent with the 90.83% RAVDESS accuracy.



(a) Accumulative magnitude

(b) Center-frequency histogram

Figure 18: Reduced filter-bank analysis after fine-tuning (RAVDESS eval features).



(a) t-SNE (RAVDESS, balanced)

(b) Quality factor $Q{=}f_c/B$

Figure 19: Reduced after fine-tuning: RAVDESS feature separability and Q-profile.

10 Discussion of Discrepancies

10.1 Why TIMIT results align closely

TIMIT is clean, studio-quality speech with limited acoustic variability. Under these conditions, even a 50-epoch schedule with random chunk sampling is sufficient for AFM frontends to approach the asymptotic accuracy of the paper. The residual differences of a few tenths of a percent are consistent with (i) single-run noise versus three-run averaging and (ii) incomplete convergence relative to 500 epochs.

10.2 Why RAVDESS and VoxCeleb gaps are larger

Out-of-domain corpora stress robustness to emotion (RAVDESS), channel, and speaker diversity (VoxCeleb). We identify five primary factors:

1. **Fine-tuning duration.** The paper adapts for 200 epochs; we use 20 ($10\times$ shorter). Domain adaptation for unconstrained Free-CNN appears especially sensitive to this budget (largest deficits in Tables 5–6).

2. **Fine-tuning corpus scale.** `train-clean-360` covers ≈921 speakers; `dev-clean` covers ≈40. The reduced speaker diversity limits the transferable gender cues acquired during fine-tuning.

3. **VoxCeleb evaluation set.** Paper numbers are for full VoxCeleb1; ours are for the official test partition only. Absolute accuracy is therefore not strictly comparable, though relative ranking among frontends remains informative.

4. **Single-run reporting.** Published tables average three seeds; we report one seed (42). Run-to-run variance of order 0.1–0.5 pp is plausible on TIMIT and larger on OOD sets.

5. **Stochastic training coverage.** Eight random chunks per utterance per epoch under-samples long utterances relative to exhaustive 10 ms hopping, which may slightly weaken low-SNR or boundary-frame robustness.

10.3 Positive observations

Despite these constraints, AFM frontends remain competitive:

- Sinc2 and Reduced recover within 1.2–2.2 pp of the paper on VoxCeleb1-test;
- Gauss slightly exceeds the paper on RAVDESS;
- Reduced achieves the best TIMIT CER with only 26 frozen filters, supporting the paper's claim that pole clustering yields a compact yet accurate bank.

11 Conclusion

We have reproduced the AFM-frontend speaker gender recognition pipeline of Fayyazi and Shekofteh (2023), spanning TIMIT training, Cascade pole reduction, LibriSpeech fine-tuning, and OOD evaluation on RAVDESS and VoxCeleb1-test. In-domain TIMIT results closely match the published means. Out-of-domain accuracy is lower, for reasons that are largely attributable to controlled protocol differences (epoch budget, fine-tuning data scale, and evaluation subset) rather than to an architectural failure of the AFM approach. The accompanying figures—training curves, complex-plane pole reduction, and filter interpretability analyses—document that the Reduced bank preserves discriminative spectral coverage with far fewer parameters, in line with the paper's central thesis.

Future work. Closing the OOD gap would require (i) fine-tuning on LibriSpeech `train-clean-360` for $\geq$200 epochs, (ii) evaluation on full VoxCeleb1, and (iii) three independent runs with mean$\pm$std reporting as in the original study.

References

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[3] J. S. Garofolo et al., “TIMIT acoustic-phonetic continuous speech corpus,” LDC93S1, Linguistic Data Consortium, 1993.

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[6] A. Nagrani, J. S. Chung, and A. Zisserman, “VoxCeleb: A large-scale speaker identification dataset,” in *Proc. Interspeech*, 2017, pp. 2616–2620.

A Additional TIMIT Training Curves

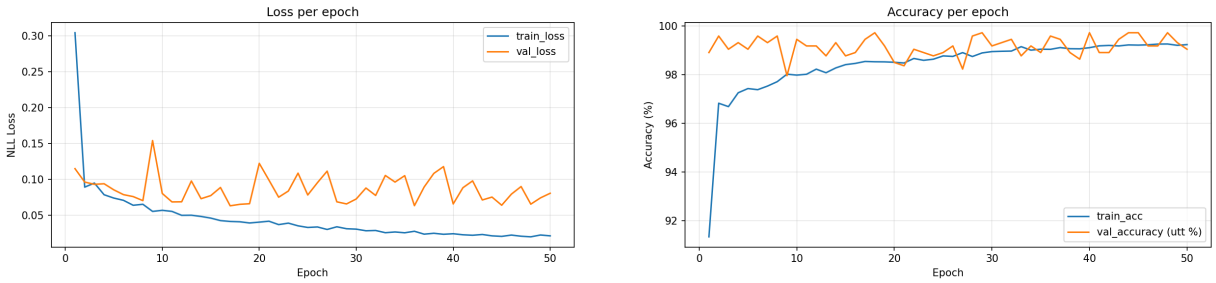


Figure 20: Sinc — TIMIT training.

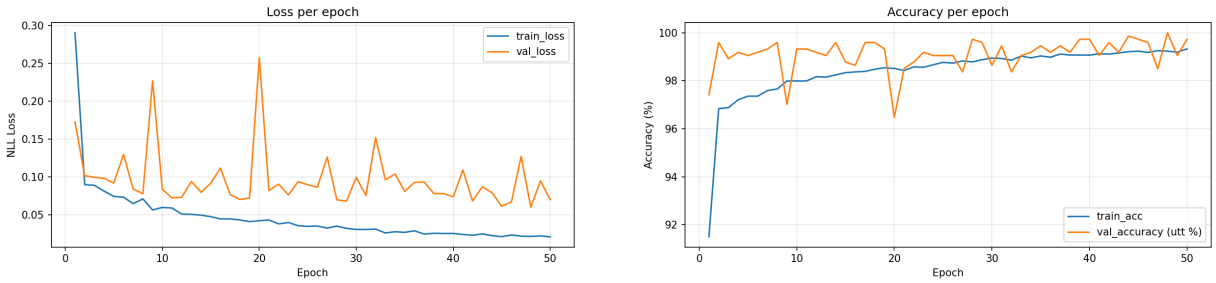


Figure 21: Sinc2 — TIMIT training.

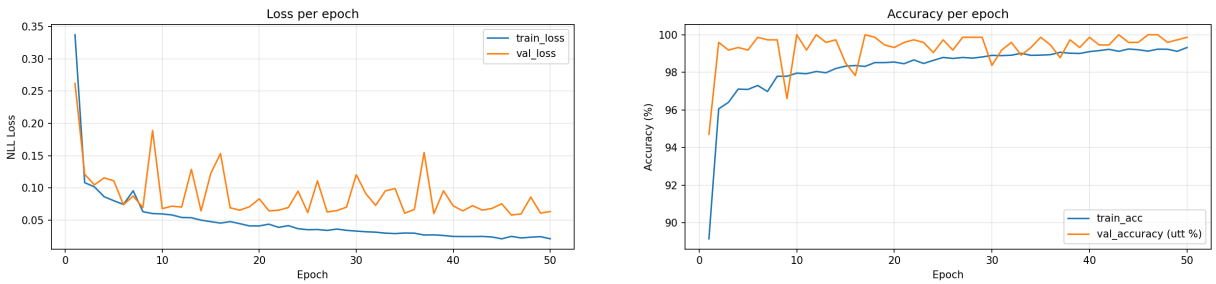


Figure 22: Gamma — TIMIT training.

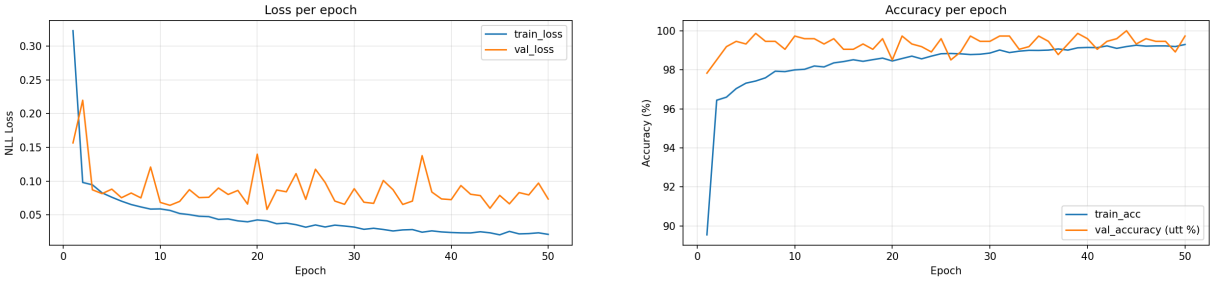


Figure 23: Gauss — TIMIT training.

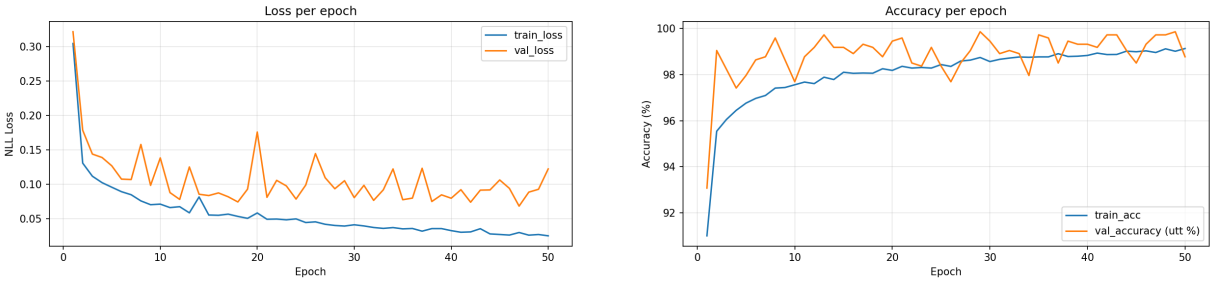


Figure 24: Free-CNN — TIMIT training.